Instruction Format Specification

IA-32 Architecture • Two-Pass Assembler

PROJECT DETAILS

Team Name: AsmCore
Repository: `github.com/SnehaDCodes/asmCore.git`
Target Arch: x86 / IA-32 (32-bit execution mode)

TEAM MEMBERS

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1. Introduction

This specification outlines the essential IA-32 machine-code encodings tailored for a two-pass assembler implementation, focusing exclusively on 32-bit registers and operand types. It details the structure of variable-length instructions and their constituent encoding fields, specifically:

- **Opcode:** Primary instruction identifier byte(s).
- **ModR/M & SIB Bytes:** Addressing mode, register specifiers, and scaled-index-base modifiers.
- **Displacements & Immediates:** Explicit 32-bit address offsets and immediate data constants.

2. IA-32 Instruction Byte Diagram

OPCODE	ModR/M	SIB	DISPLACEMENT	IMMEDIATE
Mandatory (1–2 Bytes)	Optional (1 Byte)	Optional (1 Byte)	Optional (32-bit)	Optional (32-bit)

Instruction Encoding Layout: Opcode → ModR/M → SIB → Displacement → Immediate

3. Encoding Field Meaning

Field	Meaning / Specification
`ModR/M`	Contains MOD, REG, and R/M fields; identifies 32-bit register or memory operands.
`SIB`	Scale-Index-Base byte used for complex 32-bit indexed memory addressing.
`+rd`	Opcode-embedded register addition used by **`B8+rd`** and similar families.
`id`	32-bit immediate data constant (4 bytes, little-endian).
`cd`	32-bit relative displacement / branch offset (4 bytes).
`disp32`	32-bit explicit memory address displacement (4 bytes).

4. 32-bit Register Encoding Table

Register	Code
EAX	000
ECX	001
EDX	010
EBX	011
ESP	100
EBP	101
ESI	110
EDI	111

5. Common IA-32 Opcode / Encoding Table (32-bit Only)

Instruction	Operands	Opcode	ModR/M	Mod	Reg (3–5)	R/M	+rd	SIB	Imm / Disp
MOV	r/m32, r32	89 /r	Required	variable	register code	variable	—	If required	—
MOV	r32, r/m32	8B /r	Required	variable	register code	variable	—	If required	—
MOV	r/m32, imm32	C7 /0 id	Required	variable	000	variable	—	If required	id
MOV	r32, imm32	B8 + rd id	No	—	—	—	Yes	No	id
ADD	r/m32, r32	01 /r	Required	variable	register code	variable	—	If required	—
ADD	r32, r/m32	03 /r	Required	variable	register code	variable	—	If required	—
ADD	r/m32, imm32	81 /0 id	Required	variable	000	variable	—	If required	id
ADD	EAX, imm32	05 id	No	—	—	—	—	No	id
SUB	r/m32, r32	29 /r	Required	variable	register code	variable	—	If required	—
SUB	r32, r/m32	2B /r	Required	variable	register code	variable	—	If required	—
SUB	r/m32, imm32	81 /5 id	Required	variable	101	variable	—	If required	id
SUB	EAX, imm32	2D id	No	—	—	—	—	No	id
AND	r/m32, r32	21 /r	Required	variable	register code	variable	—	If required	—
AND	r32, r/m32	23 /r	Required	variable	register code	variable	—	If required	—
AND	r/m32, imm32	81 /4 id	Required	variable	100	variable	—	If required	id
OR	r/m32, r32	09 /r	Required	variable	register code	variable	—	If required	—
OR	r32, r/m32	0B /r	Required	variable	register code	variable	—	If required	—
OR	r/m32, imm32	81 /1 id	Required	variable	001	variable	—	If required	id
XOR	r/m32, r32	31 /r	Required	variable	register code	variable	—	If required	—
XOR	r32, r/m32	33 /r	Required	variable	register code	variable	—	If required	—
XOR	r/m32, imm32	81 /6 id	Required	variable	110	variable	—	If required	id
CMP	r/m32, r32	39 /r	Required	variable	register code	variable	—	If required	—
CMP	r32, r/m32	3B /r	Required	variable	register code	variable	—	If required	—
CMP	r/m32, imm32	81 /7 id	Required	variable	111	variable	—	If required	id
CMP	EAX, imm32	3D id	No	—	—	—	—	No	id
NOT	r/m32	F7 /2	Required	variable	010	variable	—	If required	—
MUL	r/m32	F7 /4	Required	variable	100	variable	—	If required	—
IMUL	r/m32	F7 /5	Required	variable	101	variable	—	If required	—
IMUL	r32, r/m32	0F AF /r	Required	variable	register code	variable	—	If required	—
IMUL	r32, r/m32, imm32	69 /r id	Required	variable	register code	variable	—	If required	id
DIV	r/m32	F7 /6	Required	variable	110	variable	—	If required	—
INC	r/m32	FF /0	Required	variable	000	variable	—	If required	—
INC	r32	40 + rd	No	—	—	—	Yes	No	—
DEC	r/m32	FF /1	Required	variable	001	variable	—	If required	—
DEC	r32	48 + rd	No	—	—	—	Yes	No	—
JMP	rel32	E9 cd	No	—	—	—	—	No	cd
JMP	r/m32	FF /4	Required	variable	100	variable	—	If required	—
JZ	rel32	0F 84 cd	No	—	—	—	—	No	cd
JNZ	rel32	0F 85 cd	No	—	—	—	—	No	cd
CALL	rel32	E8 cd	No	—	—	—	—	No	cd
CALL	r/m32	FF /2	Required	variable	010	variable	—	If required	—
RET	—	C3	No	—	—	—	—	No	—